

Warshall's Algorithm - Transitive Closure Questions

- 1. Given the relation R on set $A = \{1,2,3\}$ with adjacency matrix: $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ Use Warshall's algorithm to find the transitive closure of R .
- 2. For the relation R on $A = \{a,b,c,d\}$ with adjacency matrix: $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ Apply Warshall's algorithm to compute the transitive closure.
- 3. Consider the relation on $A = \{1,2,3,4\}$ given by adjacency matrix: $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ Find the transitive closure using Warshall's algorithm.
- 4. For relation R on set $A = \{1,2,3\}$ with adjacency matrix: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Determine the transitive closure using Warshall's algorithm.
- 5. Given the relation on $A = \{a,b,c\}$ with adjacency matrix: $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ Apply Warshall's algorithm to find the transitive closure.
- 6. Consider the relation on $A = \{1,2,3,4\}$ represented by: $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ Use Warshall's algorithm to compute the transitive closure.
- 7. Let R be the relation on $A = \{1,2,3\}$ with adjacency matrix: $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ Find the transitive closure using Warshall's algorithm.
- 8. For the relation R on $A = \{a,b,c,d\}$ given by adjacency matrix: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ Compute the transitive closure using Warshall's algorithm.
- 9. Consider the relation on $A = \{1,2,3,4\}$ with adjacency matrix: $\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ Apply Warshall's algorithm to determine the transitive closure.
- 10. Given the relation R on $A = \{1,2,3\}$ represented by adjacency matrix: $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ Find the transitive closure using Warshall's algorithm.